

Data Modeling and Star Schemas

Document ID	DOC007
Project	Data Modeling
Category	Data Modeling
Batch	batch_2

Purpose: Original educational notes created as a searchable PDF corpus for a portfolio-scale incremental RAG pipeline. The material is designed to contain meaningful technical sections that naturally produce multiple retrieval chunks.

Purpose of Data Modeling

Data modeling defines how data entities, attributes, relationships, and business concepts are represented. Transactional models prioritize consistent operational updates, while analytical models prioritize understandable and efficient querying. A good model communicates grain, keys, relationships, and the meaning of measures.

Practical note. A model should be explainable in business terms: what one row represents, how keys relate tables, which fields are descriptive, and which values are measurable.

Facts and Dimensions

A fact table stores measurable business events at a declared grain, such as one row per order line. Measures might include quantity, sales amount, discount, or cost. Dimension tables contain descriptive context such as customer, product, date, seller, or region. Fact tables typically reference dimensions through keys.

Practical note. A model should be explainable in business terms: what one row represents, how keys relate tables, which fields are descriptive, and which values are measurable.

Star Schema

A star schema places a central fact table around denormalized dimensions. It is popular for analytics because relationships are easy to understand and queries often require fewer joins than highly normalized models. Before building a star schema, the designer should declare the fact grain and identify dimensions and measures.

Practical note. A model should be explainable in business terms: what one row represents, how keys relate tables, which fields are descriptive, and which values are measurable.

Surrogate and Natural Keys

A natural key comes from the source business domain, such as a product code. A surrogate key is generated specifically for the analytical model. Surrogate keys can isolate the warehouse from changing source identifiers and support historical dimension techniques. The source natural key is still usually retained for traceability.

Practical note. A model should be explainable in business terms: what one row represents, how keys relate tables, which fields are descriptive, and which values are measurable.

Slowly Changing Dimensions

Slowly changing dimensions describe approaches for handling changes to dimension attributes. Type 1 overwrites an old value and does not preserve history. Type 2 creates a new dimension row for a changed version and typically uses effective dates, an active indicator, or similar fields to preserve history.

Practical note. A model should be explainable in business terms: what one row represents, how keys relate tables, which fields are descriptive, and which values are measurable.

Data Model Validation

Model validation should verify uniqueness of dimension keys, expected fact grain, referential integrity, valid measures, and relationships between facts and dimensions. A technically valid table can still be wrong if its grain is unclear or joins create duplicated measures.

Practical note. A model should be explainable in business terms: what one row represents, how keys relate tables, which fields are descriptive, and which values are measurable.

End of document.